

Weekly meeting progress

Friday, Jul 17, 2026





Deployment and reliability testing of the distributed system

- Fixed RPC communication between the central router and local agents
- First deployment of the architecture on the CloudLab infrastructure
- First execution of the qDINA benchmark on CloudLab



First benchmark results (10GB DB)

- With 5 replicates, 1 100 queries and TPC-H database with 10 GB
- Total runtime: 1799.838s (qDINA: 4000s, DINA: 3500s)
- With 203 indexes:
 - Replica 0: 51
 - Replica 1: 49
 - Replica 2: 51
 - Replica 3: 35
 - Replica 4: 17



Agent's storage budget adjustment

- Agent should respect budget limit, 5 GB for 10 GB database

Two solutions:

- Stop the episode when an agent exceeds the limit (what about the other agents and router?)
- Give the agent a penalty when it exceeds the limit



Second benchmark results after fixing storage limit to 5 GB

- With 4 replicates
- Total runtime: 2536.198s (DINA and qDINA: 3000s)
- With 17 indexes:
 - Replica 0: 6
 - Replica 1: 7
 - Replica 2: 2
 - Replica 3: 2



Next tasks

- Improving agent memory management and finding what is the best solution between stopping the episode or just giving a penalty.
- Run the benchmark on the same machine as Sam on CloudLab